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Participant data synchronization in videoconferencing

Abstract

Systems, methods, and media are provided for processing participant data in a videoconference. The method includes recording an audio signal with a microphone associated with a participant of the conference and determining if the audio signal includes speech produced by the participant using a speech detection module. The method also includes determining if a time delay exists between initiation of the speech and the determination that the audio signal includes the speech produced by the participant. The method further includes adjusting a first speed of the audio signal to an adjusted first speed if the time delay is determined to exist and outputting the audio signal at the adjusted first speed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10743107	12/2019	Yoshioka	N/A	H04R 29/006
2014/0078938	12/2013	Lachapelle	370/260	A61B 17/3211
2016/0006986	12/2015	Settipalli	348/14.08	H04N 7/155
2017/0236532	12/2016	Reynolds	704/248	H04M 3/42221
2020/0092422	12/2019	Dickins	N/A	H04M 3/42221
2020/0106708	12/2019	Sleeve	N/A	H04L 65/80
2020/0349953	12/2019	Qin	N/A	G10L 15/26
2020/0403818	12/2019	Daredia	N/A	G10L 17/00
2021/0174791	12/2020	Shen	N/A	G06N 3/04
2021/0399911	12/2020	Jorasch	N/A	H04L 12/1818
2022/0303502	12/2021	Fisher	N/A	H04L 67/104
2022/0368554	12/2021	Williams	N/A	H04L 12/1831
2023/0063454	12/2022	Delsordo	N/A	H04N 5/04
2024/0406622	12/2023	Fernandez	N/A	G10L 25/30

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Background/Summary

BACKGROUND

(1) Videoconferencing systems typically connect people at a videoconferencing endpoint, such as a videoconference room, with people at other videoconferencing endpoints. During some videoconferences, participants manually mute and un-mute their microphones to prevent unwanted noise from being transmitted to the other endpoints of the videoconference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an example schematic of a videoconferencing system, according to some examples of the present disclosure.
- (2) FIG. 2 is a block diagram of a data processing system, according to some examples of the present disclosure.
- (3) FIG. 3 is a block diagram of the videoconferencing system of FIG. 1.
- (4) FIG. 4 is an example screen of a graphical user interface for configuring speech detection settings, according to some examples of the present disclosure.

(5) FIG. 5 is a time series chart for adjusting a speed of an audio signal using the videoconferencing system of FIG. 1.

(6) FIG. 6 is a flowchart of a method of implementing the videoconferencing system of FIG. 1.

(7) FIG. 7 is a block diagram of another videoconferencing system, according to some examples of the present disclosure.

(8) FIG. 8 is a flowchart of a method of implementing the videoconferencing system of FIG. 7.

(9) FIG. 9 is a schematic of an example codec, according to some examples of the present disclosure.

DETAILED DESCRIPTION

(10) Videoconferencing systems typically connect people at a videoconferencing endpoint, such as a videoconference room, with people at other videoconferencing endpoints, such as a remote work room. In such systems, audio and visual data of each participant can be recorded by microphones and cameras, respectively, before being transmitted to other videoconferencing endpoints, thereby improving communication and interaction in the videoconference.

(11) When a microphone connected to a videoconference system is used in a public place, a large conference room, or another room with two or more participants, background sounds, side conversations, or other distracting noises may be present in the audio signal that the microphone records and outputs to other participants in the videoconference. For example, a videoconference participant who is working remotely may unintentionally transmit ambient noise, e.g., side conversations, construction noise, animal sounds, to the other videoconferencing endpoints, which can lead to an undesirable videoconferencing experience for the other videoconference participants. In addition, no adequate industry standard or specification has been developed to reduce unwanted visuals, e.g., persons or objects, that enter a field of view (FOV) of a camera connected to the videoconference system from being transmitted to remote endpoints in a videoconferencing system.

(12) More specifically, objects or persons not participating in the videoconference may move into the FOV of a camera used to record visual data for the videoconference system. This may be particularly true when a participant is working remotely in a public space or with others present in the participant's vicinity. As a result, images of non-participants may be transmitted to a far end of the videoconference, which in turn may be confusing to the actual participants in the videoconference. Correspondingly, it is undesirable to transmit a particular view to a far end of the videoconference if a participant's face is not fully visible in that particular view, e.g., if the participant turns to talk to a person that is not in the FOV of the camera or gets up to walk around.

(13) Moreover, existing videoconferencing systems often rely on the participant to manually toggle the mute button of a microphone and the camera power button of a camera to reduce unwanted audio and/or video data from being transmitted to the other endpoints of the videoconference.

However, manually muting a microphone and/or turning off a camera can be error-prone, cumbersome, and may cause participants to question if their audio and video data is being received by the other participants. For example, microphone and/or camera settings from a previous videoconference may carry over into a new videoconference, which may be undesirable if a participant begins speaking while their microphone and/or camera is turned off. To prevent such errors, participants are burdened with having to pay close attention to their videoconferencing settings and manually adjust such settings throughout the videoconference as necessary. This in turn can distract participants during a videoconference and detract from conversational efficiency. Further, no adequate industry standard or specification has been developed to synchronize a participant's audio data, e.g., sound produced by a participant that is picked up by a microphone, with the participant's video data, e.g., video of the participant recorded by a camera.

(14) For many applications, it is useful to determine when a participant is speaking to other participants in the videoconference to provide for a more comprehensive understanding of the videoconference. For example, an active speaker's video stream may be highlighted or expanded on a display to draw attention to the active speaker. The ability to recognize the speech of a participant

can be enabled by using machine learning-based speech detection estimation models, but such approaches impose significant hardware and/or processing costs without providing the accuracy for differentiating participant speech that is intended to be heard by others from participant speech that is not, e.g., a side conversation with a person off-camera. Further, such approaches are limited by the time they require to process audio signals, which can cause a delay to exist between the real-time audio that is being produced by the participant and the audio that is output to the remote endpoints of the videoconference. The processing time needed to detect speech in the audio signal can also create a delay between the processed audio and a corresponding video signal, which in turn can cause confusion in the videoconference.

(15) Accordingly, in some examples, the present disclosure provides systems, methods, and media for processing and outputting data in a teleconference or a videoconference. In particular, the present disclosure provides methods of determining if a participant is speaking by analyzing audio and/or video data that is recorded by the participant's microphone and/or camera, respectively. The methods further include determining if a time delay exists between the processed data and the real-time speech of the participant, and adjusting a speed of the processed data to quickly eliminate such a time delay at a conference endpoint. Thus, a participant's microphone may be unmuted for the duration of the videoconference, though the other endpoints of the conference can be provided with a respective audio signal from the participant only when the participant is speaking. By utilizing the disclosed data processing methods, communication between participants in the videoconference may be clearer, and the overall videoconferencing experience may be more enjoyable for the participants. Further, the methods discussed herein are applicable to a wide variety of different locations and room designs, meaning that the disclosed methods may be easily assembled and applied to any particular videoconferencing or teleconferencing setting.

(16) By way of example, FIG. 1 illustrates an aspect of a videoconferencing system **10** that includes a participant **12** using a device **14** to participate in a videoconference. In some aspects, the participant **12** may be using the device **14** at a remote endpoint of the videoconference to interact with additional participants **16** at other remote endpoints of the videoconference. In some examples, the device **14** is a computer as illustrated in FIG. 1, although it is contemplated that a variety of different devices can be used to participate in a videoconference, such as, e.g., mobile phones, landline phones, smart displays, conference room equipment, and the like. Further, a microphone, i.e., a microphone array **18**, and a webcam or camera **20** can be coupled to the device **14**. In some aspects, the microphone array **18** is also configured as a speaker array, meaning that the microphone array **18** can receive and output audio signals. While FIG. 1 illustrates the microphone array **18** and the camera **20** as being integral with the device **14**, it will be understood that the microphone array **18** and/or the camera **20** can be provided as separate devices in other arrangements.

(17) The microphone array **18** can capture sound, e.g., an audio signal **24**, that is produced by the participant **12** and provide the audio signal **24** to a data processing system (DPS) **26** that can also be coupled to the device **14** or provided as software stored within and executed by the device **14**, as will be discussed below in greater detail. In a similar way, the camera **20** can capture a video signal of the participant **12** and provide the video signal to the DPS **26**. The DPS **26** can then provide the audio signal **24** and/or the video signal to one or more output devices that are located at endpoints of the videoconference, thereby allowing the participant **12** to interact with the other participants **16** in the videoconference.

(18) Generally, when a participant **12** keeps the microphone array **18** unmuted or the camera **20** on, distracting noises and/or visuals may also be transmitted to the remote endpoints of the videoconference, if, for example, a background noise is picked up by the microphone array **18** and/or another person walks within the FOV of the camera **20**. According to some examples, however, the DPS **26** can apply the processes described herein to filter the audio and/or video data while the microphone array **18** remains unmuted and/or when the camera **20** remains turned on,

which in turn can prevent distracting audio or video from being transmitted to the other endpoints of the videoconference. That is, the DPS **26** analyzes audio and/or visual data to transmit such data only when participant speech is detected. Furthermore, the DPS **26** can temporarily speed up the audio and/or video data output to a far end of the video conference in order to account for any delay in transmitting such audio data as a result of the speech detection processes to better maintain real-time communication during the videoconference.

(19) FIG. 2 illustrates an example schematic block diagram of a processing circuit **28** of the DPS **26**. In some aspects, the processing circuit **28** can include a microphone mute-controlling module **32**, an audio pre-recording module **36**, a speech detection module **38**, and/or an audio output-controlling module **40**. The processing circuit **28** can receive an audio signal from an input **34** and transmit an audio signal via an output **42**. For example, in some aspects, the input **34** includes one or more microphones, e.g., a microphone array, to capture audio produced by the participant and provide the audio signal to the processing circuit **28**. The microphones may be of any suitable type, such as, e.g., dynamic, condenser, electret, ribbon, carbon, piezoelectric, fiber optic, laser, omnidirectional, directional, and/or MEMS type. In some aspects, the microphones can be built-in microphones in a videoconferencing device such as the device **14** (see FIG. 1), or the microphones can be separate devices, e.g., headset microphones, webcam microphones, Bluetooth microphones, etc., that are coupled to the DPS **26**.

(20) The mute-controlling module **32** can be used to automatically un-mute a participant's microphone when a participant enters a videoconference, thereby allowing the participant's microphone to stay un-muted for the duration of the videoconference. This can advantageously prevent errors that may occur when manually muting and un-muting a microphone to selectively transmit audio to endpoints of the videoconference, as will be discussed in greater detail below. In some examples, the mute-controlling module **32** can defer to manual muting and un-muting of the microphone, meaning that the processing circuit **28** allows a participant to override the mute-controlling module **32**, if desired. Thus, absent manual control by the participant, the microphone mute-controlling module **32** can normally un-mute the microphone for the duration of a videoconference.

(21) Referring still to FIG. 2, the audio signal obtained at the input **34** can be provided to the pre-recording module **36**, which can store the audio signal for further processing. In some aspects, the pre-recording module **36** can be an audio buffer, meaning that the pre-recording module **36** may temporarily store a digital audio signal to prevent audio loss or audio clipping. In some examples, the pre-recording module **36** may be configured to store a first segment, e.g., a predetermined time period, of an audio signal. For example, the first segment of the audio signal may include the first ten, or the first five, or the first three words spoken by the participant, or the first segment of the audio segment may include about the first 10 seconds of the audio signal, or about the first five seconds of the audio signal, or about the first two seconds of the audio signal, or about the first one second of the audio signal. Thus, it is contemplated that any discussion herein of a pre-recorded audio signal may also be applicable to a first segment of an audio signal, as will be discussed below in greater detail. In this way, the combination of the mute-controlling module **32** and the pre-recording module **36** can ensure that all audio produced by the participant is recorded, which in turn decreases speech recording delay and preserves conversational fluidity in the videoconference.

(22) The speech detection module **38** can determine if the audio signal contains speech produced by the participant by accessing the audio signal stored in the pre-recording module **36**. The speech detection module **38** may detect speech using a variety of processes, such as, e.g., embedding/feature extraction, pattern matching, signal power-level detection, etc. In some aspects, the speech detection module **38** can identify speech in the audio signal using an artificial intelligence (AI) machine learning model such as, e.g., hidden Markov models, deep learning models, convolution neural networks, recurrent neural networks, etc. Such methods can be used alone or in combination and, in some aspects, may be selected by the participant through a settings

portal of the video conference or the participant's device.

(23) With continued reference to FIG. 2, using a speech detection process, the speech detection module **38** can determine that the audio signal includes speech produced by the participant, which in turn can cause the audio output-controlling module **40** to output the recorded audio signal to an output **42**. In some aspects, the output **42** can be at a remote end audio device, such as, e.g., a loudspeaker, a personal computer, a headset, a mobile phone, etc. Thus, the audio output-controlling module **40** can selectively output an audio signal to remote ends of a videoconference based on the determination that the audio signal includes participant speech. If the determination is negative, that is, if the speech detection module **38** does not detect participant speech in the audio signal, the audio output-controlling module **40** may not pass the audio signal to the output **42**. In this way, the processing circuit **28** can prevent distracting audio from being transmitted to remote ends of the videoconference, even if the input **34**, remains unmuted during the videoconference. Correspondingly, the processing circuit **28** eliminates the need to manually mute and unmute the input **34**, which in turn can allow a participant to stay actively engaged in the videoconference.

(24) Furthermore, in some aspects, the audio output-controlling module **40** can detect if a time delay exists between the audio signal that is recorded by the input **34** and the audio signal that is provided from the audio output controlling module **40** to the output **42**. Such a time delay may be created as the recorded audio signal is processed along the processing circuit **28**. For example, the speech detection module **38** may perform speech recognition for a predetermined time period, e.g., a few seconds, before determining that the recorded audio signal contains speech produced by the participant.

(25) If such a time delay does exist, the audio output-controlling module **40** can adjust an output speed of the audio signal to gradually align the audio signal output to the output **42** with the real-time speed at which the audio signal is being produced at the input **34**. Specifically, the audio output-controlling module **40** can access the pre-recording module **36** and adjust a speed of the pre-recorded audio signal stored therein to gradually minimize or substantially eliminate the time delay. For example, the audio output-controlling module **40** may determine that a time delay exists, e.g., if the audio signal to the output **42** is significantly delayed from the real-time audio signal being produced by the participant, and the audio output-controlling module **40** can accelerate the pre-recorded audio signal to gradually eliminate such a time delay. Without such acceleration, the audio signal transmitted to far ends of the video conference may be significantly delayed, disrupting the videoconference and/or causing a discrepancy between the audio and video signals heard and seen at the far end (e.g., the participant's speech may not match up with their lip movement in their video). Alternatively, without pre-recording the audio data and temporarily accelerating the output audio signal, the beginning of the participant's audio may simply be cut off and not transmitted to the far end of the video conference, disrupting the videoconference and/or causing a discrepancy between the audio and video signals heard and seen at the far end.

(26) Referring now to FIG. 3, an example block diagram is illustrated of the videoconferencing system **10**, which can include an input **34**, the DPS **26**, an output **42**, and a web-conferencing service **44**. While the aspects of the videoconferencing system **10** are described as relating to a single participant, i.e., a single remote endpoint of a videoconference, it will be understood that the systems and methods herein are applicable each remote endpoint in a videoconference and to additional videoconferencing arrangements, e.g., a conference room with multiple participants. As discussed above, the input **34** can be used to record participant data that is then processed using the DPS **26**. For example, the input **34** can include a microphone **46** and/or a camera **48** to record audio and/or visual data, respectively. While the input **34** is depicted in FIG. 3 as a single component including the microphone **46** and the camera **48**, it should be noted that the system **10** may include multiple separate inputs **34** in some applications. Additionally, in some applications, the input **34** may only include a microphone **46**. For example, in one application, the system **10** may be a teleconferencing system that only processes and transmits audio signals.

(27) Referring still to FIG. 3, the input **34** can be coupled to the DPS **26**, and the DPS **26** can include the microphone mute-controlling module **32**, the pre-recording module **36**, the speech detection module **38**, and the audio output-controlling module **40**. As described above with respect to FIG. 2, the microphone mute-controlling module **32** can be used to automatically un-mute the microphone **46** when the participant joins the videoconference, although it is contemplated that the participant may override the mute-controlling module **32** by manually muting/un-muting the microphone if desired. When un-muted, the microphone **46** can record an audio signal, and the audio signal can be stored in the pre-recording module **36** as a digital pre-recorded audio signal. The audio signal is then provided to the speech detection module **38** which applies a speech detection technique, as discussed above, to detect if the audio signal does or does not contain speech produced by the participant. In some aspects, the speech detection module **38** can differentiate the participant's speech from speech produced by other persons. For example, the speech detection module **38** may include one or more learning models that are trained to recognize a particular person's speech patterns.

(28) In some examples, an installer or participant can manually adjust the settings of the speech detection module **38** using, for example, a graphical user interface (GUI) on a display of a device. For example, FIG. 4 illustrates an example screen with a graphical user interface (GUI) **50** for receiving user input to configure speech detection settings. It is contemplated that the GUI may be accessible in a variety of different ways, such as, e.g., under a settings tab in a videoconferencing software application, under settings in an audio driver, visible in gaming software user interfaces, and/or under general computer settings. In some aspects, the GUI **50** includes a menu of speech detection settings **52** that can be manually changed before, during, and/or after a videoconference. Further, a variety of different inputs may be used to allow a participant to control certain aspects in the GUI **50**, including any acceptable human interface devices, e.g., touch enabled devices, button inputs, keyboards, mice, track balls, joysticks, touch pads, and/or the like. The speech detection settings **52** may be provided in any suitable format, such as drop-down menus (DDLs), field boxes, selection boxes, etc. In the non-limiting example illustrated in FIG. 4, the speech detection settings **52** are provided as selection boxes, including at least a “speech detection by microphone” selection box **54** and a “speech detection by camera” selection box **56**. Accordingly, a participant can customize the speech detection settings **52** to suit the participant's environment. For example, the participant may select the “speech detection by microphone” selection box **54** and de-select the “speech detection by camera” selection box **56** if the participant is in range of the microphone **46** but not in view of the camera **48**, e.g., if the participant is standing up or walking around during the videoconference.

(29) Further, the speech detection settings **52** may include one or more sub-settings under the selection boxes **54**, **56** that can be selected to change the method of speech detection. For example, the GUI **50** can include a “Method #1” selection box **58** and a “Method #2” selection box **60** positioned under the “speech detection by microphone” selection box **54**, where “Method #1” and “Method #2” each correspond to different audio speech detection processes, e.g., short-time detection, AI model detection, power level detection, etc. Correspondingly, the GUI **50** can include sub-settings under the “speech detection by camera” selection box **56**, such as, e.g., an “Eye tracking” selection box **62**, a “Mouth tracking” selection box **64**, and/or a “Head pose tracking” selection box **66**. Thus, the speech detection module **38** (shown in FIG. 3) can be configured to detect speech from an audio signal, a video signal, or both via a variety of different selectable methods. Further, it is contemplated that the GUI **50** may include settings or options other than the speech detection settings **52** illustrated in FIG. 4.

(30) Returning to FIG. 3, the speech detection module **38** can provide the audio output-controlling module **40** with a determination after performing speech detection on the data recorded by the input **34**. If speech is found to exist in the recorded audio signal, the audio output-controlling module **40** can prepare the audio signal to be output to remote endpoints of the videoconference. If the speech

detection module **38** does not detect speech in the audio signal, the audio output-controlling module **40** may not transmit the audio signal to remote endpoints of the videoconference. In this way, the audio output-controlling module **40** can automatically prevent sounds other than participant speech, e.g., distracting background noises, from being transmitted to other participants in the videoconference, even if the microphone **46** remains un-muted for the duration of the videoconference. In other words, the audio output-controlling module **40** achieves the effect of manually muting the microphone **46** while also eliminating the need for the participant to constantly pay attention to the status of the microphone **46**, which in turn can allow the participant to focus on the subject matter of the videoconference and lead to a more cohesive videoconferencing experience.

(31) Additionally, the audio output-controlling module **40** can adjust an output speed of the audio signal to reduce or eliminate any time delay that exists therein, e.g., a time delay that is created between the received, real-time audio signal and the pre-recorded audio signal as a result of processing the pre-recorded audio signal in the DPS **26**. To that end, the audio output-controlling module **40** can include a synchronization module **68** to determine if a time delay exists, a speed adjustment module **70** to adjust a speed of the pre-recorded audio signal stored in the pre-recording module **36**, and an output module **72** to transmit the adjusted audio signal from the DPS **26**.

(32) Referring now to FIG. 5, an example time series chart **74** is shown to demonstrate how the output speed of the audio signal is adjusted using the audio output-controlling module **40**. Specifically, the chart **74** illustrates that the audio signal can be recorded at a first timestamp **T0**, meaning that the microphone **46** (see FIG. 3) can detect and receive the audio signal and the pre-recording module **36** can begin storing the audio signal at the first timestamp **T0**. The speech detection module **38** is then utilized to determine if the audio signal includes speech produced by the participant, a second timestamp **T1** can be generated upon detecting participant speech in the audio signal, resulting in a determination to output the audio signal to a far end of the videoconference. Further, the synchronization module **68** can compare the first timestamp **T0** (e.g., time at which the audio signal is received) with the second timestamp **T1** (e.g., time at which it is determined to output such audio signal) to determine if a significant time delay exists therebetween. For example, the synchronization module **68** may determine that the second timestamp **T1** occurs later than the first timestamp **T0**, meaning that a time delay, e.g., a first time period **76**, exists between determining that the pre-recorded audio signal should be output and the true speed of the audio signal. Upon such a determination, the synchronization module **68** can calculate the speed at which the pre-recorded audio must be output to eliminate the time delay **76** by a future third timestamp **T2**. However, an additional time delay, e.g., a second time period **78**, may exist between the second timestamp **T1** and the third timestamp **T2**, so the synchronization module **68** may calculate the total time delay to be the sum of the first time period **76** and the second time period **78**.

(33) After the synchronization module **68** performs the above determination, the speed adjustment module **70** can access the pre-recording module **36** and adjust the output speed of the pre-recorded audio signal. For example, if the synchronization module **68** determines that the second timestamp **T1** occurs later than the first timestamp **T0** by a threshold amount, the speed adjustment module **70** can accelerate an output speed of the pre-recorded audio signal and output the accelerated audio signal at the second timestamp **T1**. Specifically, the pre-recorded audio signal can be accelerated to align with the real-time audio signal, e.g., the real-time signal that is being produced by the participant, at or before the third timestamp **T2**. Put another way, the pre-recorded audio signal can be accessed at the second timestamp **T1**, accelerated through the second time period **78** until reaching the third timestamp **T2**, when it has caught up with the real-time audio signal and may again be output at a normal (i.e., unaccelerated) rate. As discussed above, the pre-recorded audio signal may include a first segment of an audio signal, meaning that an output speed of a first segment of an audio signal can also be adjusted by the speed adjustment module **70**.

(34) Further, the speed adjustment module **70** may be a dynamic module, meaning that the speed adjustment module **70** can adjust the output speed according to a variety of dynamic factors, such as, e.g., delay time, speech rate, processing settings, silent audio clips, etc. Thus, in some examples, the speed adjustment module **70** may dynamically adjust the speed of the pre-recorded audio signal proportionally with respect to a size of the time delay, meaning that greater acceleration can be used to eliminate larger time delays. For example, the speed adjustment module **70** may accelerate the pre-recorded audio at a speed that is 1.2 times faster than the real-time audio signal speed to eliminate a first time delay, and/or the speed adjustment module **70** may accelerate the pre-recorded audio at a speed that is 1.1 times faster than the real-time audio signal speed to eliminate a second time delay that is less than the first time delay. Alternatively, if no time delay exists, e.g., if the first timestamp **T0** and the second timestamp **T1** occur simultaneously or substantially simultaneously (e.g., the time delay is nominal, below a threshold amount), the audio output-controlling module **40** may not adjust the speed of the pre-recorded audio signal. Accordingly, it is an advantage of the disclosure that the audio output-controlling module **40** can selectively adjust the speed of a pre-recorded audio signal to reduce and/or eliminate any time delay that may be created as the pre-recorded audio signal is processed in the DPS **26**. In this manner, a participant at a videoconferencing end point will not notice any delay in or clipping of the speaking participant's speech as a result of the processing time required by the DPS **26** for speech detection.

(35) Referring back to FIG. **3**, the audio signal can be transmitted from the DPS **26** at the adjusted speed using the output module **72**. In some aspects, the adjusted audio signal is transmitted to a web-conferencing service **44**, such as, e.g., Zoom, Microsoft Teams, Cisco Webex, Google Meet, etc., before then being transmitted to the output **42** at a conference endpoint. In other examples, the output module **72** may transmit the audio signal directly to the output **42**. As discussed above, for example, the output **42** can include a speaker **80** and/or a display **82** to output audio and/or video data, respectively. In particular, the speaker **80** can be used to output the audio signal at the adjusted first speed as a result of the processing performed by the DPS **26**, as discussed above. While the output **42** is depicted in FIG. **3** as a single component including the speaker **80** and the display **82**, it should be noted that the system **10** may include multiple separate outputs **42** in some applications.

(36) In light of the above, FIG. **6** illustrates an example method **100** of implementing the videoconferencing system **10**. In some aspects, the steps of the method **100** can be stored as instructions stored in memory and carried out via modules of the DPS **26** described above. Furthermore, while the steps of FIG. **6** are illustrated in a particular order, in some aspects, the steps may be carried out in a different order, simultaneously, and/or more or fewer steps may be carried out.

(37) Referring still to FIG. **6**, at step **102**, the method **100** can include automatically unmuting a microphone that is associated with a participant or user by using a microphone mute-controlling module when the user joins a videoconference. As discussed above, the user can optionally override the microphone mute-controlling module by manually muting the microphone. Further, the microphone mute-controlling module can be in communication with and/or connected to a monitor and/or a codec that includes a memory and a processor, as will be discussed below in greater detail. At step **104**, the method **100** includes recording an audio signal with the microphone and then providing or transmitting the audio signal to a pre-recording module, e.g., a pre-recording buffer. The pre-recording buffer may be configured to store the audio signal or at least a segment of the audio signal. At step **106**, the method **100** can include determining if the audio signal recorded by the microphone includes speech produced by the user and, if the audio signal is determined not to include user speech, returning to step **104**. However, if the audio signal is determined to include user speech at step **106**, the method **100** can include additional processing steps.

(38) Specifically, the method **100** can proceed to step **108**, in which timestamps corresponding to the time at which the audio signal is first recorded by the microphone and the time at which speech

is detected in the audio signal are compared. Put another way, a first timestamp can be generated upon initially recording the audio signal, and a second timestamp can be generated upon determining that the audio signal includes speech produced by the user. At step **110**, the method **100** can include determining if a time delay exists between the first timestamp and the second timestamp. That is, step **110** can include determining if the first timestamp occurs earlier than the second timestamp. If such a delay is determined to exist, e.g., if the difference between the second timestamp and the first timestamp exceeds a threshold amount, the method **100** can include step **112** of adjusting a first speed of the audio signal to gradually reduce or eliminate the time delay. For example, the output-controlling module can be used to accelerate the pre-recorded audio signal relative to the real-time speed at which the audio signal is produced if the second timestamp occurs later than the first timestamp. It is contemplated that the speed of the audio signal can be accelerated at any rate suitable to reduce the time delay, e.g., 1.1×, 1.2×, 1.5×, 2×, etc.

(39) After the speed of the audio signal is adjusted, the method **100** can include step **114** of outputting the audio signal at the adjusted speed. In this way, the output audio can gradually align with the real-time speed at which the audio signal is being produced, which in turn can reduce audio delay and lead to a more desirable videoconferencing experience. Alternatively, if no time delay is determined to exist at step **110**, the method **100** can proceed directly to outputting the unaltered audio signal at step **114**. After outputting the audio signal, the method **100** can return to step **104** and repeat steps **104**, **106**, **108**, **110**, **112**, and/or **114** to continue recording, processing, and outputting audio during the videoconference. As a result, the method **100** can determine when the output audio signal has “caught up” with the input audio signal, and speed adjustments to the audio signal no longer need to be made.

(40) Generally, the method **100** can be performed in real-time or near real-time. For example, in some aspects, the steps **102**, **104**, **106**, **108**, **110**, **112**, and **114** of the method **100** are repeated after a period of time has elapsed, such as, e.g., at least every 30 seconds, or at least every 15 seconds, or at least every 10 seconds, or at least every 5 seconds, or at least every 3 seconds, or at least every second, or at least every 0.5 seconds. It should be noted that the above method **100**, or any methods or processes described herein, can be implemented as a set of instructions, tangibly embodied on a non-transitory computer-readable media, such that a processor device can implement the instructions based upon reading the instructions from the computer-readable media.

(41) The above methods and systems generally relate to processing and adjusting audio data. However, such systems and methods may equally apply to video data in some aspects. For example, FIG. 7 illustrates an example block diagram of another videoconferencing system **200**, according to some aspects. It will be understood that any components or aspects of the videoconferencing system **200** may also be compatible with the videoconferencing system **10** illustrated in FIG. 3, and that the videoconferencing systems **10**, **200** may be combined in a single videoconferencing system, as will be discussed below in greater detail. In general, the videoconferencing system **200** may be configured to record, process, and output a video signal in a similar way to how the videoconferencing system **10** records, processes, and outputs an audio signal to prevent distracting data from being transmitted to remote endpoints of a videoconference. Thus, elements of the videoconferencing system **200** that are shared with, i.e., that are structurally and/or functionally identical or similar to, elements present in the videoconferencing system **10** are represented by like reference numerals in the **200** series.

(42) In the non-limiting example illustrated in FIG. 7, the videoconferencing system **200** can include a DPS **226**, a camera power-controlling module **232**, an input **234**, a pre-recording video module **236**, a speech detection module **238**, a video output-controlling module **240**, and a web-conferencing service **244**. Similar to the input **34** of FIG. 3, the input **234** can include one or more input devices, such as, e.g., a microphone **246** and a camera **248** to record audio and/or video data, respectively. Further, the power-controlling module **232** can be used to automatically power on the camera **248** when the participant joins the videoconference, although it is contemplated that the

participant may override the power-controlling module **232** by manually powering off the camera **248** if desired.

(43) The video signal recorded by the camera **248** can be stored in the pre-recording video module **236** as a pre-recorded video signal. In some aspects, the pre-recording video module **236** is configured to store a first segment, e.g., a predetermined time period, of a video signal. For example, the first segment of the video signal may include the first 30, or the first 15, or the first 5 frames of the video signal, or the first segment of the video signal may include about the first 10 seconds of the video signal, or about the first five seconds of the video signal, or about the first two seconds of the video signal, or about the first second of the video signal. Thus, it is contemplated that any discussion herein of a pre-recorded video signal may also be applicable to a first video segment of a video signal, as will be discussed below in greater detail.

(44) Further, the speech detection module **238** can be used to determine if the participant is speaking by applying a speech detection process to input audio and/or video signals, similar to the speech detection module **38** of FIG. **3** described above. If the speech detection module **238** confirms that the participant is speaking, the video output-controlling module **240** may access the pre-recording video module **236** and, if necessary, adjust the pre-recorded video signal to improve the videoconferencing experience.

(45) With continued reference to FIG. **7**, the video output-controlling module **240** can include a synchronization module **268**, a speed adjustment module **270**, and an output module **272**. The synchronization module **268** can be used to determine if a first time delay exists between the pre-recorded video signal being received and a time of speech detection. For example, the synchronization module **268** can compare a first timestamp, e.g., a timestamp generated upon initiation of video recording, with a second timestamp, e.g., a timestamp generated upon determining that the participant is speaking. If the second timestamp is different or occurs later than the first timestamp, e.g., if a first time delay exists between the first and second timestamps, the synchronization module **268** can calculate the speed at which the pre-recorded video signal can be output to gradually reduce and/or eliminate the time delay. Additionally, in some aspects, the synchronization module **268** can also determine if a second time delay exists between the pre-recorded video signal and a pre-recorded audio signal. For example, a fourth timestamp may be generated when sound is first recorded by the microphone **246**, and the synchronization module **268** can compare the first timestamp with the fourth timestamp to determine if a second time delay exists therebetween. Upon such a determination, the synchronization module **268** may calculate speeds at which both the pre-recorded audio signal and the pre-recorded video signal may be output to gradually reduce and/or the second time delay. This in turn can lead to a more synchronized combination of audio and video data being output to remote ends of the videoconference.

(46) If the synchronization module **268** detects the first time delay and/or the second time delay, the speed adjustment module **270** can access the pre-recording video module **236** and adjust the output speed of the pre-recorded video signal. For example, the speed adjustment module **270** can accelerate a first speed of the video signal if the first timestamp associated with the initiation of video recording occurs before the second timestamp associated with positive speech recognition. By accelerating the video signal, the speed adjustment module **270** can reduce video latency, leading to a smoother video being output to remote endpoints of the videoconference in real time.

(47) Further, the speed adjustment module **270** may also accelerate or decelerate the video signal and/or an audio signal if the first timestamp associated with the initiation of video recording is different than the fourth timestamp associated with the initiation of audio recording. For example, the speed adjustment module **270** may decelerate the first speed of the video signal, accelerate the second speed of the audio signal, or both to eliminate the second time delay. Accordingly, it will be understood that the speed adjustment module **270** can adjust speeds of audio and/or video signals recorded by the input **234** to provide aligned data streams to remote endpoints in a videoconference. However, the speed adjustment module **270** may leave audio and/or video signals

unaltered if no time delay is determined to exist or if a time delay exists below a pre-defined threshold.

(48) With continued reference to FIG. 7, the video signal can be transmitted from the DPS **226** at the adjusted speed using the output module **272**. In some aspects, the adjusted audio signal is transmitted to a web-conferencing service **244**, such as, e.g., Zoom, Microsoft Teams, Cisco Webex, Google Meet, etc., before then being transmitted to the output **242**, or the output module **272** may transmit the audio signal directly to the output **242**. As discussed above, the output **242** can include output devices that are located at endpoints of the videoconference, such as a speaker **280** and a display **282** to output audio and/or visual data, respectively. In particular, the display **282** can be used to output the video signal at the adjusted first speed as a result of the processing performed by the DPS **226**, as discussed above.

(49) FIG. **8** illustrates a method **300** of implementing the videoconferencing systems **10**, **200** discussed above. At steps **302** and **304**, the method **300** can include automatically unmuting a microphone and powering on a camera, respectively, that are associated with a participant when the participant joins a videoconference. At step **306**, the method **300** can include recording and storing data associated with the participant in a pre-recording module or buffer. In some examples, the user data can include audio and video signals, which are recorded with the microphone and the camera, respectively. Further, the pre-recording buffer may be configured to store the user data or at least segments of the user data. At step **308**, the method **300** can include determining if the user is speaking in the videoconference, e.g., using audio and/or visual speech detection techniques. If the user is determined not to be speaking in the videoconference, the method can return to step **306**, such that no audio or video data is output to far ends of the video conference. However, in some applications, video data may still be output though audio data is not output following such determination at step **308**.

(50) If the user is determined to be speaking at step **308**, the method **300** may proceed to step **310**, which can include comparing timestamps associated with the user data, the timestamps being generated at various points along the method **300**. Specifically, at step **310**, the method **300** can include comparing timestamps corresponding to the times at which the user data is first recorded by the microphone and/or the camera, and the time at which the user is determined to be speaking. For example, a first timestamp can be generated upon initially recording the audio signal, a second timestamp can be generated upon initially recording the video signal, and a third timestamp can be generated upon determining that the user is speaking in the videoconference. At step **312**, the method **300** can include determining if any time delays exist in the user data, e.g., time delays between any combination of the timestamps discussed above. That is, step **312** includes determining if the first timestamp is different, e.g., occurs earlier or later, than the second timestamp and/or the third timestamp. Additionally, step **312** can include determining if the second timestamp is different e.g., occurs earlier or later, than the first timestamp and/or the third timestamp.

(51) If a time delay is determined to exist, the method **300** can include step **314** of adjusting a first speed of the participant data relative to a real-time speed at which the participant data is being produced to gradually reduce or eliminate the time delay and/or synchronize audio and video data. For example, the output-controlling module can be used to accelerate the pre-recorded audio signal and/or the pre-recorded video signal relative to a real-time speed at which such signals are produced if the first timestamp and/or the second timestamp, respectively, are determined to occur earlier than the third timestamp. Correspondingly, the output-controlling module can be used to adjust the speeds of the pre-recorded audio signal and/or the pre-recorded video signal relative to one another if a time delay exists between the first timestamp and the second timestamp. For example, the output-controlling module may accelerate a first speed of the pre-recorded audio signal, accelerate a second speed of the pre-recorded video signal, or both if the first timestamp is determined to have occurred later than the second timestamp. Thus, it is contemplated that the

speed of the participant data can be adjusted at any rate suitable to reduce the time delay, e.g., 1.1×, 1.2×, 1.5×, 2×, -1.1×, -1.2×, -1.5×, -2×, etc. In one example, audio data can be accelerated while video data may be decelerated (i.e., adjusted by a factor less than one) so that the output appears more synchronized and less visually disruptive to a user at the far end of the video conference until the participant data catches up.

(52) Further, the method **300** can include step **316** of outputting the participant data, or normal speed if no time delays exist, or at the adjusted speed if time delays exist. In this way, the output audio and video data can gradually align with the real-time speed at which the audio signal and video signals are being produced, which in turn will reduce audio and/or video delay and lead to a more desirable videoconferencing experience. As noted above, if no time delay is determined to exist at step **312**, the method **300** can proceed directly to outputting the unaltered user data at step **316**. After outputting the audio signal, the method **300** can return to step **306** and repeat steps **306**, **308**, **310**, **312**, **314**, and/or **316** to continue recording, processing, and outputting user data during the videoconference.

(53) Generally, the method **300** can be performed in real-time or near real-time. For example, in some aspects, the steps **302**, **304**, **306**, **308**, **310**, **312**, **314**, and **316** of the method **300** are repeated after a period of time has elapsed, such as, e.g., at least every 30 seconds, or at least every 35 seconds, or at least every 30 seconds, or at least every 5 seconds, or at least every 3 seconds, or at least every second, or at least every 0.5 seconds. It should be noted that the above method **300**, or any methods or processes described herein, can be implemented as a set of instructions, tangibly embodied on a non-transitory computer-readable media, such that a processor device can implement the instructions based upon reading the instructions from the computer-readable media.

(54) FIG. **9** illustrates aspects of a codec **400** according to some examples of the present disclosure. In some aspects, a codec **400** may be a separate device of a videoconferencing system or may be integrated within a participant's videoconferencing device, e.g., a computer, a mobile phone, a tablet, etc. Generally, the codec **400** includes machine readable instructions to maintain a video call with one or more videoconferencing end points, receive audio and video streams from microphones and cameras, respectively, and encode and composite the streams, according to the methods described herein, to send to the one or more remote end points. In some applications, components of the videoconferencing systems **10**, **200** described above may be incorporated into or coupled to the codec **400**.

(55) As shown in FIG. **9**, the codec **400** may include loudspeaker(s) **402**, though in many cases the loudspeaker **402** is provided in a display **404**. The codec **400** may include microphone(s) **406** interfaced via a bus **408**, connected through an analog to digital (A/D) converter **410**, and the loudspeaker **402** is connected through a digital to analog (D/A) converter **412**. The codec **400** also includes a processing unit **414**, a network interface **416**, a flash or other non-transitory memory **418**, RAM **420**, and an input/output (I/O) general interface **422**, all coupled by the bus **408**. A camera **424** is connected to the I/O general interface **422**. Additional microphone(s) **406** are connected to the network interface **416**. An HDMI interface **426** is connected to the bus **408** and to the external display or monitor **404**. Bus **408** is illustrative and any interconnect between the elements can be used, such as Peripheral Component Interconnect Express (PCIe) links and switches, Universal Serial Bus (USB) links and hubs, and combinations thereof. The camera **424** and microphones **406**, **406** can be contained in housings containing the other components or can be external and removable, connected by wired or wireless connections.

(56) The processing unit **414** can include digital signal processors (DPSs), central processing units (CPUs), graphics processing units (GPUs), dedicated hardware elements, such as neural network accelerators and hardware codecs. The flash memory **418** stores modules of varying functionality in the form of software and firmware, generically programs or machine readable instructions, for controlling the codec **400**. Illustrated modules include a video codec **428**, camera control **430**, framing **432**, other video processing **434**, audio codec **436**, audio processing **438**, network

operations **440**, user interface **442** and operating system, and various other modules **444**. In some examples, a power controlling module, a pre-recording module, a speech detection module, and/or an output-controlling module, as discussed above, are included with the modules included in the flash memory **418** and can be executed by the processing unit **414**. Furthermore, in some examples, machine readable instructions can be stored in the flash memory **418** that cause the processing unit **414** to carry out steps of any of the methods **100**, **300** described above. The RAM **420** is used for storing any of the modules in the flash memory **418** when the module is executing, storing video images of video streams and audio samples of audio streams and can be used for scratchpad operation of the processing unit **414**.

(57) The network interface **416** enables communications between the codec **400** and other devices and can be wired, wireless or a combination. In one example, the network interface **416** is connected or coupled to the Internet **446** to communicate with remote endpoints **448** in a videoconference. In one example, the general interface **422** provides data transmission with local devices (not shown) such as a keyboard, mouse, printer, projector, display, external loudspeakers, additional cameras, and microphone pods, etc.

(58) In one example of this disclosure, the processing unit **414** processes the video and audio from the camera **424** and the microphones **406** using processes in the modules stored in the flash memory **418**. Processed audio and video streams can be sent to and received from remote devices coupled to network interface **416** and devices coupled to general interface **422**. Microphones in a microphone array used for speech recognition can be used as the microphones providing speech to the far site, or separate microphones, such as microphone **406**, can be used.

(59) Certain operations of methods according to the technology, or of systems executing those methods, can be represented schematically in the figures or otherwise discussed herein. Unless otherwise specified or limited, representation in the figures of particular operations in particular spatial order can not necessarily require those operations to be executed in a particular sequence corresponding to the particular spatial order. Correspondingly, certain operations represented in the figures, or otherwise disclosed herein, can be executed in different orders than are expressly illustrated or described, as appropriate for particular examples of the technology. Further, in some examples, certain operations can be executed in parallel, including by dedicated parallel processing devices, or separate computing devices that interoperate as part of a large system.

(60) The disclosed technology is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. Other examples of the disclosed technology are possible and examples described and/or illustrated here are capable of being practiced or of being carried out in various ways.

(61) A plurality of hardware and software-based devices, as well as a plurality of different structural components can be used to implement the disclosed technology. In addition, examples of the disclosed technology can include hardware, software, and electronic components or modules that, for purposes of discussion, can be illustrated and described as if the majority of the components were implemented solely in hardware. However, in one example, the electronic based aspects of the disclosed technology can be implemented in software (for example, stored on non-transitory computer-readable medium) executable by a processor. Although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes. In some examples, the illustrated components can be combined or divided into separate software, firmware, hardware, or combinations thereof. As one example, instead of being located within and performed by a single electronic processor, logic and processing can be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components can be located on the same computing device or can be distributed among different computing devices connected by a network or other suitable communication links.

(62) Any suitable non-transitory computer usable or computer readable medium may be utilized.

The computer-usable or computer-readable medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device. More specific examples (a non-exhaustive list) of the computer-readable medium would include the following: a portable computer diskette, a hard disk, a random-access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, or a magnetic storage device. In the context of this disclosure, a computer-usable or computer-readable medium may be any medium that can contain, store, communicate, or transport the program for use by or in connection with the instruction execution system, apparatus, or device.

(63) As used herein in the context of computer implementation, unless otherwise specified or limited, the terms “component,” “system,” “module,” “block,” and the like are intended to encompass part or all of computer-related systems that include hardware, software, a combination of hardware and software, or software in execution. For example, a component can be, but is not limited to being, a processor device, a process being executed (or executable) by a processor device, an object, an executable, a thread of execution, a computer program, or a computer. By way of illustration, both an application running on a computer and the computer can be a component. Components (or system, module, and so on) can reside within a process or thread of execution, can be localized on one computer, can be distributed between two or more computers or other processor devices, or can be included within another component (or system, module, and so on).

Claims

1. A method of processing participant data in a conference, the method comprising: recording an audio signal with a microphone associated with a participant of the conference; determining if the audio signal includes speech produced by the participant using a speech detection module; determining if a time delay exists between when the participant initiates the speech and the determination that the audio signal includes the speech produced by the participant; adjusting a first speed of the audio signal to an adjusted first speed if the time delay is determined to exist; and outputting the audio signal at the adjusted first speed.
2. The method of claim 1, the method further comprising: automatically unmuting the microphone with a microphone mute-controlling module when the participant joins the conference.
3. The method of claim 1, wherein recording the audio signal includes storing a first segment of the audio signal in a pre-recording buffer, and wherein adjusting the first speed of the audio signal includes accessing the first segment of the audio signal that is stored in the pre-recording buffer.
4. The method of claim 3, wherein adjusting the first speed of the audio signal further includes adjusting the first segment of the audio signal.
5. The method of claim 1, wherein recording the audio signal occurs at a first timestamp, and wherein using the speech detection module occurs at a second timestamp, and wherein determining if the time delay exists includes comparing the first timestamp with the second timestamp to determine if the second timestamp is different than the first timestamp.
6. The method of claim 5, wherein adjusting the first speed of the audio signal includes accelerating the first speed of the audio signal to eliminate the time delay when the second timestamp occurs later than the first timestamp.
7. The method of claim 1, the method further comprising: recording a video signal with a webcam; determining if an additional time delay exists between the audio signal and the video signal; adjusting a second speed of the video signal if the additional time delay is determined to exist; and outputting the video signal at the adjusted second speed.
8. The method of claim 1, the method further comprising: recording a video signal with a webcam; adjusting a second speed of the video signal by a factor less than one; and outputting the video signal at the adjusted second speed.

9. A non-transitory computer-readable medium containing instructions that, when executed, cause a processor to: record an audio signal with a microphone associated with a participant of a conference; determine if the audio signal includes speech produced by the participant using a speech detection module; determine if a time delay exists between when the participant initiates the speech and the determination that the audio signal includes the speech produced by the participant; adjust a first speed of the audio signal to an adjusted first speed if the time delay is determined to exist; and output the audio signal at the adjusted first speed.
 10. The non-transitory computer-readable medium of claim 9, wherein the audio signal includes a first audio segment that is stored in a pre-recording buffer, the first audio segment having a second speed.
 11. The non-transitory computer-readable medium of claim 10, wherein the second speed of the first audio segment is adjusted if the time delay is determined to exist.
 12. The non-transitory computer-readable medium of claim 9, wherein the processor is to record a first timestamp when the audio signal is recorded and a second timestamp when the audio signal is determined to include the speech produced by the participant.
 13. The non-transitory computer-readable medium of claim 12, wherein the first speed of the audio signal is accelerated to eliminate the time delay if the second timestamp occurs later than the first timestamp.
 14. The non-transitory computer-readable medium of claim 9, wherein the processor is to record a video signal with a webcam and determine if a time delay exists between the audio signal and the video signal.
 15. A system for processing data in a videoconference, the system comprising: a microphone associated with a participant of the videoconference to record an audio signal; a camera associated with the participant to record a video signal; a processor connected to the microphone and the camera; and a memory coupled to the processor, the memory including: a pre-recording module to store the audio signal and the video signal; a speech detection module to determine if the participant is speaking in the videoconference; and an output-controlling module to adjust at least one of a first speed of the audio signal and a second speed of the video signal if a time delay is determined to exist between the audio signal and the video signal, wherein the memory stores a first timestamp that is generated when the audio signal is recorded; a second timestamp that is generated when the video signal is recorded; and a third timestamp that is generated when the participant is determined to be speaking in the videoconference.
 16. The system of claim 15, wherein the memory further includes a microphone mute controlling module to unmute the microphone.
 17. The system of claim 15, wherein the speech detection module applies at least one of an audio speech detection process and an artificial intelligence motion tracking process to determine if the participant is speaking.
 18. The system of claim 15, wherein the output-controlling module accelerates the first speed of the audio signal if the third timestamp occurs later than the first timestamp, and wherein the output-controlling module accelerates the second speed of the video signal if the third timestamp occurs later than the second timestamp.
 19. The system of claim 15, wherein the output-controlling module accelerates the first speed of the audio signal, deaccelerates the second speed of the video signal, or both to eliminate the time delay between the audio signal and the video signal if the first timestamp occurs later than the second timestamp.
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